CHAPTER 14

Nanoparticle-impregnated biopolymers as novel antimicrobial nanofilms

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14.1 Introduction

Recent attention towards developing biocompatible, biodegradable, and bioactive films for diverse applications in food packaging, therapeutics, and textiles has led to the exploration of biopolymers and the fabrication of novel nanocomposites. Biopolymer-based packaging materials act as barriers to moisture, water vapor, gases, and solutes, and thus can improve food quality, extending shelf life through minimizing microbial growth in or on the product. Furthermore, biopolymer films are excellent vehicles for incorporating a wide variety of additives, such as antioxidants, antifungal agents, antimicrobials, colors, and other nutrients. Thus, nanofilms composed of biopolymeric matrices are being intensely explored for their applications in material science, electronics, and biomedical science (Rhim and Ng, 2007). Biopolymer nanofilms incorporated with nanoparticles display markedly improved mechanical, thermal, optical, and physicochemical properties compared to pure polymers. Moreover, rational incorporation of antimicrobial peptides, antibiotics, or nanoparticles may inhibit the adherence of bacteria on the surface and thereby inhibit biofilm formation.

Microbial contamination is associated with microbes' ability to adhere and colonize various animate and inanimate surfaces forming biofilms. Microbial communities universally attach to surfaces and produce various extracellular polysaccharides, DNA, and proteins owing to which they form biofilms on wound dressings, industrial pipes, food packages, and separation membranes, which has become a serious global concern (Lu et al., 2015). Numerous pathogenic bacteria can produce biofilms on silicon, latex, polycarbonate, stainless steel, glass, and polyvinyl chloride, which are commonly used in food processes, storage, and reconstitution equipment (Iversen et al., 2006). One of the primary causes of nosocomial infection is bacterial adhesion followed by biofilm formation onto medical devices like catheters, implants, and surgical instruments. These matrix-enclosed bacterial populations are highly resistant to antibiotics and other biocidal agents compared to their planktonic counterparts (Ghosh et al., 2015a,b,c,d).

Biopolymer-based Nano Films
https://doi.org/10.1016/B978-0-12-823381-8.00017-X

The development of biofilms on the surfaces of medical implants and devices not only compromises their functionality but also limits their lifetime. Biofilm-associated infections on surfaces of tissues (such as teeth, skin, and the urinary tract) are critical health hazards. Biofilms are responsible for several diseases such as otitis media (acute ear infection), bacterial endocarditis, cystic fibrosis, Legionnaire's disease (acute respiratory infection), kidney stones, urinary tract infections, osteomyelitis, periodontal diseases, and nosocomial infections (Costerton et al., 1999; Davies, 2003; Römling and Balsalobre, 2012). Biofilms associated with scratched surfaces of improperly cleaned or sanitized equipment in the dairy industry leads to contamination of milk and milk products (Srey et al., 2013). An increase in microbial cell populations on the surface leads to the formation of a biofilm, which consists of a polysaccharide matrix with embedded cells that enables microbial cells to survive even under unfavorable harsh conditions. Moreover, embedded cells exhibit a thousand times reduced susceptibility to antimicrobial agents like antibiotics and other biocides. Further, various toxins excreted from biofilm-associated cells markedly increase the pathogenicity and resilient spreading of infections. Additionally, antibiotic resistant gene transfer between closely packed biofilm-associated bacteria enhances the emergence of multidrug-resistant bacterial strains (Siedenbiedel and Tiller, 2012).

Recently, novel antibacterial surfaces for direct killing of bacteria upon contact, and/or reduction in bacterial attachment and subsequent biofilm formation, have gained considerable attention. Such materials either can actively or passively prevent biofilm formation. Active "bacteria-killing" materials include cationic biopolymers, antimicrobial peptides, antibiotics, silver ions, nitric oxide, and others, whereas passively "bacteria-resistive" materials are comprised of polyethylene glycol (PEG), zwitterionic polymers, and their derivatives (Cheng et al., 2007; Saldarriaga et al., 2007;Cheng et al., 2009). However, both active and passive materials have been reported to eradicate a biofilm only partially, which emphasizes the growing need to integrate these approaches to design high-performance antibacterial surfaces that can strongly resist bacterial adhesion and/or kill bacteria, thereby preventing biofilm formation. Nanomaterials with a large surface area and size-/shape-dependent physicochemical propertiesc play a significant role in designing antimicrobial films that might not allow biofilm formation.

Various nanostructures composed of silver, gold, copper, zinc, and graphene are being used to fabricate environmentally safe surface coatings with strong antibacterial effects and good biocompatibility (Muñoz-Bonilla and Fernandez-Garcia, 2012). Similarly, antibacterial biopolymeric membranes are also impregnated with biocidal agents, like iron oxide nanoparticles (IONPs), copper oxide nanoparticles (CuONPs), zinc oxide nanoparticles (ZnONPs), titanium dioxide nanoparticles (TiO_2NPs), graphene oxide (GO), carbon nanotubes (CNTs), and others (Mukherjee and De, 2018). Biopolymeric materials with good structure flexibility can effectively integrate antimicrobial nanostructures and form uniform surface coatings on various substrates

(Wang et al., 2008; Ilknur et al., 2010). Moreover, multi-functionalized nanomaterials impregnated in biopolymeric matrices exhibit the controlled release of antimicrobial agents for sustained bactericidal effects apart from their resistance to biofilm formation (Eby et al., 2009; Rosa and Gianfranco, 2011).

In light of this, it is a powerful and profitable strategy to combine nanoparticles and a biopolymer matrix to form multifunctional biofilm-resistant composite nanofilm coatings for antibacterial applications. Herein, we present a detailed and comprehensive account of the recent advances in the fabrication and application of biopolymer/nanoparticle composites and their promises as antibacterial biofilm-resistant surface coatings. Initially, we provide an elaborate description of the various types of antimicrobial polymer matrices followed by a discussion of their structural modification by nanoparticle impregnation. Subsequently, we present diverse mechanisms of antibiofilm activity of smart nanofilms along with their applications in various fields like healthcare and the food and textile industries.

14.2 Overview of antimicrobial biopolymeric matrices

Starch, polylactic acid (PLA), polyhydroxyalkanoate (PHA), cellulose, and chitosan are some of the inexpensive, biodegradable biopolymers used for the synthesis of nanofilms. In order to fabricate biofilm-resistant films, antimicrobial polymers are preferable as they can synergistically enhance biofilm resistance when impregnated with biocidal nanostructures. Antimicrobial polymers represent a class of biocides that can be tethered to surfaces without losing their biological activity. The first report on antimicrobial polymers was by Cornell and Dunrarumain in 1965 in which they reported polymers and copolymers prepared from 2-methacryloxy troponones to kill bacteria. These polymeric substances demonstrate superior efficacy, enhanced biocompatibility, reduced environmental hazards, and greater biofilm resistance (Siedenbiedel and Tiller, 2012). The desired feature of such surfaces is to effectively repel microbes so that they cannot attach to the surface. Similarly, these surfaces killed microbes coming in contact with them, which was achieved by loading the surfaces with biocides such as silver, antimicrobial ammonium compounds, antibiotics, active chlorine, or triclosan (Tiller, 2008). More recently, a diverse group of microbicidal nanoparticles has been impregnated for further enhancement of biofilm resistance. The following section presents an overview of recent developments in the field of antimicrobial polymers. It is important to note that critical parameters determining the extent of bactericidal action of antimicrobial polymers include molecular weight, type/degree of alkylation, and distribution of charge on the polymer. Broadly, three general types of antimicrobial polymers identified are polymeric biocides, biocidal polymers, and biocide-releasing polymers, which we discuss in the senctions that follow.

14.2.1 Polymeric biocides

Polymeric biocides are polymers that covalently link bioactive repeating units with antimicrobial activity from such chemical groups as amino, carboxyl, or hydroxyl groups. It has been reported that polymerization of the antimicrobial 4-vinyl-*N*-benzylpyridinium chloride and subsequent cross-linking resulted in a nonbiocidal water-insoluble polymer that was capable to capture microbes but unable to kill them (Kawabata, 1992). Nathan et al. showed that the antibiotics penicillin V and cephradine exhibited full antimicrobial activity when conjugated with PEG-lysine via a hydrolytically labile bond, indicating reactivation of the antibiotics upon release (Nathan et al., 1993). Copolymerization of methacrylate-modified norfloxacin and PEG-methacrylates does not lead to a loss of activity. However, reduction in the antimicrobial activity of an active polymer was noted via direct modification of vancomycin with PEG-methacrylate and subsequent polymerization. This marked decrease in activity was speculated to be proportional to the greater PEG length (Dizman et al., 2005; Lawson et al., 2009).

Further, penicillin on attachment to polyacrylate nanoparticles showed greater activity against methicillin-resistant *Staphylococcus aureus* (MRSA) than the free antibiotic in solution (Turos et al., 2007). Polymeric biocides may also contain antimicrobial substances that include hydrophobic quaternary ammonium side groups, sulfonium salts, benzimidazole, halogen, and *N*-halamine, which can successfully prevent the growth of bacteria like *Escherichia coli*, *Micrococcus luteus*, *Bacillus subtilis*, and *S. aureus* (Tashiro, 2001; Huang et al., 2016).

14.2.2 Biocidal polymers

Amphiphilic polycations are attracted to microbial cells, which possess a negative net charge at the surface due to their membrane proteins, teichoic acids of Gram-positive bacteria, and negatively charged phospholipids at the outer membrane of Gram-negative bacteria. This interaction between the cell surface and polymers leads to disruption of the outer membrane as well as the cytoplasmic membrane resulting in lysis-mediated leakage of the cellular content and eventually cell death (Timofeeva and Kleshcheva, 2011). Several antimicrobial polycations with quarternary ammonium and phosphonium, tertiary sulfonium, and guanidinium functions have been reported, as summarized in Table 14.1 (Siedenbiedel and Tiller, 2012). Poly(hexamethylene biguanidinium hydrochloride) was the first polymer to become a commercial disinfectant (Ikeda et al., 1984). Following this, poly(methacrylate) containing chlorhexidine-like side groups, polymeric mimics for the antimicrobial peptides (AMPs) like magainin and defensin, and poly (phenylene ethynylene)-based conjugated polymers with amino side groups have also been considered as potent biocidal polymers (Ikeda et al., 1986; Tew et al., 2002, 2010; Ilker et al., 2004; Gabriel et al., 2008; Lienkamp et al., 2008).

Table 14.1 Biocidal polymers for antimicrobial applications.

Polymer	Target	Remark
Quaternary ammonium polyethyleneimine	Gram-positive and Gram-negative bacteria	n-Alkylated polyethyleneimine has effective antimicrobial activity, dependent on the hydrophobic and positively charged immobilized long polymeric chains
Quaternary phosphonium modified epoxidized natural rubber	*Staphylococcus aureus*, *Escherichia coli*	Moderate growth inhibition of microbes
Arginine-tryptophan-rich peptide	Gram-positive and Gram-negative bacteria	Retain antimicrobial functionality for at least 21 days, showing negligible cytotoxicity
Guanylated polymethacrylate	*Staphylococcus epidermidis*, *Candida albicans*	Guanidine copolymers were much more active compared to the amine analogues
Chitosan	Bacteria, yeast, fungi	Widely-used antimicrobial agent either alone or blended with other compounds
Ammonium ethyl methacrylate homopolymers	Methicillin-resistant *Staphylococcus aureus*, *Escherichia coli*	Very little or no hemolytic activity and higher inhibitory effects against Gram-positive bacteria than Gram-negative bacteria
Metallo-terpyridine carboxymethyl cellulose	*Staphylococcus aureus*, *Streptococcus thermophilus*, *Escherichia coli*, *Saccharomyces cervisiae*	Minimum inhibitory concentration ranged from 6 to 8 mg/L to achieve _90% inhibition
Poly (n-vinylimidazole) modified silicone rubber	*Pseudomonas aeruginosa*, *Staphylococcus aureus*	More antibacterial activity against *Pseudomonas aeruginosa* than *Staphylococcus aureus*

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The copolymer class of dimethylaminomethyl styrene and octylstyrene is antimicrobially activated upon protonation of the tertiary amino groups, while the respective quarternized derivatives are less active (Gelman et al., 2004). Similar copolymers have been fabricated by copolymerizing dimethylaminoethyl acrylamide and aminoethyl acrylamide, respectively, with n-butylacrylamid, which is reported to have antimicrobial activity and reduced toxicity compared to the respective quarternary ammonium derivatives (Palermo and Kuroda, 2009; Palermo et al., 2009).

Poly(diallylammonium salts) containing either secondary or tertiary amino groups have been reported to be active against both *S. aureus* and *Candida albicans* (Timofeeva et al., 2009).

Dendritic and hyperbranched polymers have also been reported to exhibit strong antimicrobial properties (Pasquier et al., 2008). The cationic ring-opening polymerization strategy of 2-alkyl-1,3-oxazolines and terminating the macromolecule with a cationic surfactant helps in the controlled introduction of a specific group at one end and another group at the distal end owing to use of a suitable initiator and termination agent during synthesis. It is interesting to note that the activity of the biocidal end group, dimethyldodecylammonium bromide (DDA), is independent of the molecular weight of poly(2-methyl-1,3-oxazolines) (PMOx) and the respective ethyl-derivative (PEtOx). The satellite group, that is, the group distal to the biocide, controls the activity of the whole molecule in a range of three orders of magnitude, which is thought to be due to the amphipatic behavior of the macromolecules that play a critical role in forming unimolecular micelles in solutions and insertion of the biocide into the membrane of a microbial cell (Waschinski et al., 2005). Various related studies strongly provide evidence and endorse these facts. It was seen that a highly active PMOx derivative with an antimicrobial DDA group and a nonbioactive decyl satellite group quickly perforated an *E. coli*-like liposome membrane (Waschinski et al., 2008). However, on altering the satellite to a hexadecyl group, a loss of antimicrobial activity was noted for the whole molecule. Latter inactive macromolecules exhibited stronger binding to the liposomes than the active ones, but without penetrating the membrane, signifying the fact that the nature of the end groups strongly influences the bioactivity of antimicrobial polymers (Mowery et al., 2009; Siedenbiedel and Tiller, 2012).

14.2.3 Biocide-releasing polymers

Vogl and Tirell first polymerized a new group of antimicrobial smart materials called biocide-releasing macromolecules. This group can be realized by: (1) polymerization of biocide-releasing molecules to a polymeric backbone or (2) polymer/biocide-releasing molecular composites (Vogl and Tirrell, 1979; Huang et al., 2016). The rational incorporation of antibiotic and/or antiseptic compounds within the polymeric matrix exhibits not only antibacterial properties but also controls the release of biocidal agents, maintaining a high local biocide concentration close to microbes and facilitating the delivery of biocides with short in vivo half lives. A chitosan-agarose hybrid material and nanocomposite ionogels decorated with silver were produced using an ionic liquid and 1-butyl-3-methylimidazolium chloride; they exhibited antimicrobial activity, biocompatibility, electrical conductivity, and thermal/conformational stability (Trivedi et al., 2014). Other biocide-releasing polymers include iron-sequestering polymer-antibiotic composites, medical-grade silicone incorporated with crystal violet and di(octyl)phosphinic-acid-capped ZnONPs, and tributyltin esters of polyacrylates (Huttinger et al., 1982; El-Gendy et al., 2015; Noimark et al., 2015). Further, the attachment of *N*-halamine groups to achieve various polymeric oxidants helps in the long-term

Table 14.2 Biocide-releasing polymers with antimicrobial activity.

Sr.No.	Polymer	Target	Antimicrobial substance
1	Dextrans	*S. aureus*	Gentamicin
2	Poly-L-lysine, polyethylene glycol	*S. aureus*	Staphylolytic LysK enzyme
3	Poly(octanediol-co-citrate)	*S. aureus*, *E. coli*	Choline chloride, tetraethylammoniumbromide, hexadecyltrimethylammonium bromide, methyltriphenylphosphonium
4	Cyclodextrin	*S. aureus*, *E. coli*	Triclosan
5	Poly(methyl methacrylate)	*Pseudomonas aeruginosa*, *S. aureus*	Silver
6	Poly(methyl methacrylate)	*Staphylococcus epidermidis*, *E. coli*	Silver, nanoparticles, and imidazole complex
7	Cyclodextrin	*S. aureus*, *E. coli*	Silver, chitosan
8	Acrylic bone cements	*Enterococcus faecalis V583*	Chlorhexidrina
9	Polycaprolactone	*S. aureus*, *P. aeruginosa*	Silver

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storage of active chlorine, which may release chlorine or hypochlorite-killing microbes due to the oxidation of phospholipids on the cell membrane (Siedenbiedel and Tiller, 2012). Similarly, recent innovations like nitric oxide (NO)-releasing, natural polyphenol conjugated, photoactivated polymers are summarized in Table 14.2.

14.3 Passive or active polymers

Huang et al. (2016) reported the classification of polymeric substances based on their mode of action. Some polymers directly kill bacteria and thus resist or eradicate the microbial biofilm, while others do not allow a biofilm to grow on them due to their surface properties. The following section gives an account of two major types of polymers that are used for antimicrobial surfaces based on their mode of inhibition.

14.3.1 Passive action

A passive polymer layer avoids protein adsorption on its surface, which affects the adherence of bacteria. Microbes with mainly hydrophobic and negatively charged properties are repelled significantly by these polymeric surfaces but are not killed. The main types of passive polymers are: (1) hydrophilic, (2) negatively charged, or (3) have a low surface-free energy (Francolini et al., 2015; Zhang and Chiao, 2015). Typical passive polymers comprise: (1) self-healing, slippery liquid-infused porous surfaces (SLIPS), such as poly(dimethyl siloxane); (2) uncharged polymers, such as poly(ethylene glycol) (PEG), poly(2-methyl-2-oxazoline), polypeptoid, polypoly(n-vinyl-pyrrolidone), and poly(dimethyl acrylamide); and (3) charged polyampholytes and zwitterionic polymers, such asphosphobetaine, sulfobetaine, and phospholipid polymers (Liu et al., 2014; Ye and Zhou, 2015). Among these passive polymers, PEG has been extensively studied and has demonstrated excellent antimicrobial effects in drastically reducing protein adsorption and bacterial adhesion (Table 14.3). Due to high chain mobility, large exclusion volume, and the steric hindrance effect of a highly hydrated layer (Zhang and Chiao, 2015), PEG has been the most commonly used passive antimicrobial material (Yu et al., 2014a).

Table 14.3 Passive polymers for antimicrobial applications.

Polymer	Target	Remark
Poly(ethylene glycol)	*Staphylococcus aureus*, *Escherichia coli*, *Pseudomonas aeruginosa*	Used as neutral polymer brush systems to prevent protein and cell adhesion
Poly(sulfobetaine methacrylate)	*Pseudomonas aeruginosa*, *Staphylococcus epidermidis*	Resist protein adsorption, cell attachment, and bacterial adhesion
Poly[3-dimethyl (methacryloyloxyethyl) ammonium propane sulfonate-b-2-(diisopropylamino) ethyl methacrylate]	*Staphylococcus aureus*	Zwitterionic coronae and pH-responsive cores can impart bacterial anti-adhesive properties
Poly(2-methyl-2-oxazoline)	*Escherichia coli*	Dual-functional antimicrobial surface of poly(L-lysine)-graft-poly(2-methyl-2-oxazoline)-quarternery ammonium
Albumin, whey	*Bacillus subtilis*, *Escherichia coli*	No bacterial growth was observed on albumin-glycerol and whey-glycerol after 24 h inoculation
Polyphenols	*Streptococcus mitis*, *Fusobacterium nucleatu*, *Porphyromonas gingivalis*	Effective against periodontal bacteria

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Research has shown that it exhibits a high antifouling ability to prevent protein and cell adhesion effectively, consequently preventing the growth of microbes.

14.3.2 Active action

Active polymers effectively kill bacteria that adhere to the polymer surface. These polymers can be multifunctionalized with active agents, such as cationic biocides, antimicrobial peptides, antibiotics, or antimicrobial nanoparticles, which results in contact killing of harmful microbial pathogens. The diverse mode of killing depends on the active agents chosen to impregnate the polymeric matrix. Functionalization of the polymers with positively charged quaternary ammonium that targets the cell wall and destroys the cytoplasmic membrane results in the leakage of intracellular components, eventually leading to cell death (Xue et al., 2015). In addition, polyethylenimine, polyguanidine, and *N*-halamine are representative polymers that demonstrate active antimicrobial activity. Polyethylenimine interacts and ruptures the bacterial cell membrane, while polyguanidine inhibits bacterial growth through adhesion and subsequent disruption of Ca^{2+} salt bridges or cell death. *N*-halamine makes cell inhibition or inactivation by the action of the oxidative halogen targeted at thio or amino groups of cell receptors (Jain et al., 2014).

14.4 Nanostructures for impregnation in biopolymeric films

Spectacular advances in the field of nanotechnology have enabled the synthesis of various nanoparticles with exotic shapes and sizes. These nanoparticles have profound utility in nano-electronics, silver nanowires, nano-connectors, electrodes, single-electron transistors, information stockpiling gadgets, dynamic waveguides in optical gadgets, high thickness recording gadgets, intercalation materials for batteries, and fundamental capacitors. More recently, nanomaterials have received great attention due to their medical applications, including wound dressing, diagnosis, and pharmacological treatment. Various nanomaterials are being impregnated into polymeric films for developing biofilm-resistant surfaces. Although mechanical properties of organic polymers are not suitable for device coatings, they can be coated on metal surfaces by spin coating, dip coating, or layer-by-layer plasma polymerization. The following section gives a detailed account of various metal nanostructures incorporated into biofilm-resistant polymeric matrices.

14.4.1 Silver-based nanofilms

Silver ions and nanoparticles (AgNPs) exhibit a broad spectrum of biocidal activity against bacteria, fungi, and viruses and are the most preferred active component in silver-based antimicrobials (Ghosh and Chopade, 2017; Kale et al., 2017; Ghosh et al., 2018). Although there are several methods for the synthesis of AgNPs, biological routes employing medicinal plants like *Barleria prionitis*, *Dioscorea oppositifolia*, *Gloriosa* *superba*, *Dioscorea bulbifera*, *Platanus orientalis*, *Litchi chinensis*, and *Gnidia glauca* have gained more attention recently (Ghosh et al., 2012, 2016a,b,c; Shende et al., 2017, 2018; Shinde et al., 2018). Various techniques like co-reduction, physical mixing of preformed AgNPs with biopolymers, sputtering, and plasma deposition have been used to incorporate silver into polymeric matrixes (Sambhy et al., 2006). AgNPs were generated in polyamide 6 (PA6) by the thermal reduction of Ag^{1+} ions during the melt processing of a PA6/silver acetate mixture. Although PA6 individually does not have any antimicrobial activity, PA6/Ag nanocomposites can kill *E. coli* completely within 24 h. Retention of antimicrobial activity even after being immersed in water for 100 days indicated that PA6 filled with 2 wt% nanosilver is an effective antimicrobial material for long-term applications (Damm et al., 2007).

Further, a biopolymeric soy protein isolation (SPI) film was embedded with AgNPs by mixing both and then casting on a 5×5 cm^2 polystyrene plate followed by drying at room temperature. The dried films were immersed into a 70% (*v*/v) ethanol aqueous solution containing 1% (*v*/v) epichlorohydrin in order to cross-link at 60°C for 1 h. It is significant to note that SPI/AgNP nanofilms containing only 0.1% of Ag can also kill 90% of *E. coli* and 80% of *S. aureus* within 12 h (Zhao et al., 2013).

An innovative approach using supercritical carbon dioxide to impregnate silicone with nanoparticulate silver metal has been reported with attributed antimicrobial properties to the silicone elastomer. Bacteria adhering to such surfaces could be killed within 1 h. This impregnation strategy is advantageous as it resists bacterial colonization in both the inner and outer surfaces of catheters. Such surfaces continue to release silver ions in antimicrobial concentrations evenin the presence of a conditioning film, thus enhancing antimicrobial efficacy over a wide spectrum (Furno et al., 2004).

Bacterial cellulose from *Acetobacter xylinum* (strain TISTR 975) and *Gluconacetobacter xylinus* was impregnated with AgNPs for use as a wound dressing since it provides a moist environment resulting in better wound healing. Such freeze-dried AgNP-impregnated bacterial cellulose exhibited strong antimicrobial activity against *E. coli* (Maneerung et al., 2008; Maria et al., 2010). Advanced technological development has helped in the production of printable antibacterial devices by drop-on demand (DoD) inkjet technology, using AgNPs and polyvinyl butyral (AgNPs/PVB) as complex printable fluids and cellulose acetate as a flexible substrate (Barrera et al., 2018).

Recently, the impact of several commercial Ag dressings, namely, Acticoat (nanocrystalline silver on polyethylene mesh, NCPE), Silverlon (metallic Ag on nylon core, MSN), Aquacel (Ag silver carboxymethylcellulose, SCMC), SilverCel (metallic Ag with alginate, MSAL), and PolyMem Silver (metallic Ag with starch copolymers on a PU membrane, MSPU) were reported to reduce *P. aeruginosa* (PAO1), methicillin-resistant *S. aureus* (UC18 strain), and *E. coli* (BSC11380 strain) (Kostenko et al., 2010). Hydrophilic dressings (SCMC, MSAL, and MSPU) were reported to decrease such bacteria activity with time, unlike hydrophobic dressings (NCPE and MSN), which possessed prolonged efficacy due to a sustained Ag release at lesser amounts. AgNPs exert antimicrobial activity at 10-fold lesser concentrations than Ag ions.

The better efficacy of AgNPs seems to be due to multiple mechanisms of action: (1) the release of Ag ions; (2) the generation of reactive oxygen species (ROS); and (3) the direct damage of cell membranes (Francolini et al., 2017). Pre-treatment of central venous catheters (CVCs) with a thin polydopamine layer induced formation of AgNPs that formed a homogeneous compact coating comprised of well-separated spherical AgNPs 30–50 nm in size. This was found to be highly active against *S. aureus* (Wu et al., 2015).

In another study, AgNPs were embedded together with heparin in an *N*-vinylpyrrolidone and *N*-butyl methacrylate copolymer resulting in a bifunctional coating for CVCs with combined antimicrobial and antithrombotic properties. Antimicrobial synergy between AgNPs and heparin was observed against *S. aureus* adhesion using this coating (Stevens et al., 2011). AgNPs embedded in *Aloe vera* and curcumin containing polymethacrylic acid hydrogel-based wound dressings can resist *S. aureus* and *E. coli* colonization and promote wound healing in an animal model (Anjum et al., 2016). Nanosliver nanohydrogels (nSnH) were synthesized by nanoemulsion polymerization of methacrylic acid (MAA) followed by subsequent cross-linking and irradiation-mediated silver reduction. It is important to note that the shape, size, and formation of AgNPs were solely influenced by polymerization and irradiation time. A PVA/polyethylene oxide/carboxymethyl cellulose matrix acted as a gel system to blend with nSnH, *A. vera*, and curcumin, and as a coating on hydrolyzed polyester fabric to develop antimicrobial dressings (Fig. 14.1A and B). These dressings can release silver up to ∼42% of the total loading after 48 h.

A simple and novel technique for on-site precipitation of silver bromide (AgBr) was used to synthesize cationic polymer/AgBr nanoparticle composites with long-lasting antibacterial activity against both Gram-positive and Gram-negative bacteria. The cationic polymer (NPVP) used in this composite was poly(4-vinyl-*N*-hexylpyridinium bromide), which was embedded with AgBrNPs. These AgBrNPs serve as a source of strongly biocidal but nontoxic Ag^+ ions. This water-insoluble composite can form good coatings on glass and exhibit prolonged biocidal activity against both airborne and waterborne bacteria. It is important to note that the antibacterial activity of these composites can be significantly tuned by controlling the size of the embedded AgBrNPs. Potent antibiofilm activity was observed when glass surfaces coated with 1:1 AgBr/21% NPVP and 21% NPVP alone were incubated with *P. aeruginosa* (Sambhy et al., 2006). Surfaces without AgBrNPs were almost entirely covered with a compact bacterial biofilm after 48 h of incubation, whereas no biofilm formation was observed on 1:1 AgBr/21% NPVP-coated surfaces even after 72 h (Fig. 14.1C–F).

Advances in endodontics involve incorporation of AgNPs into matrices like root canal sealer, cements, and gutta-percha to resist recolonization of bacteria and biofilm

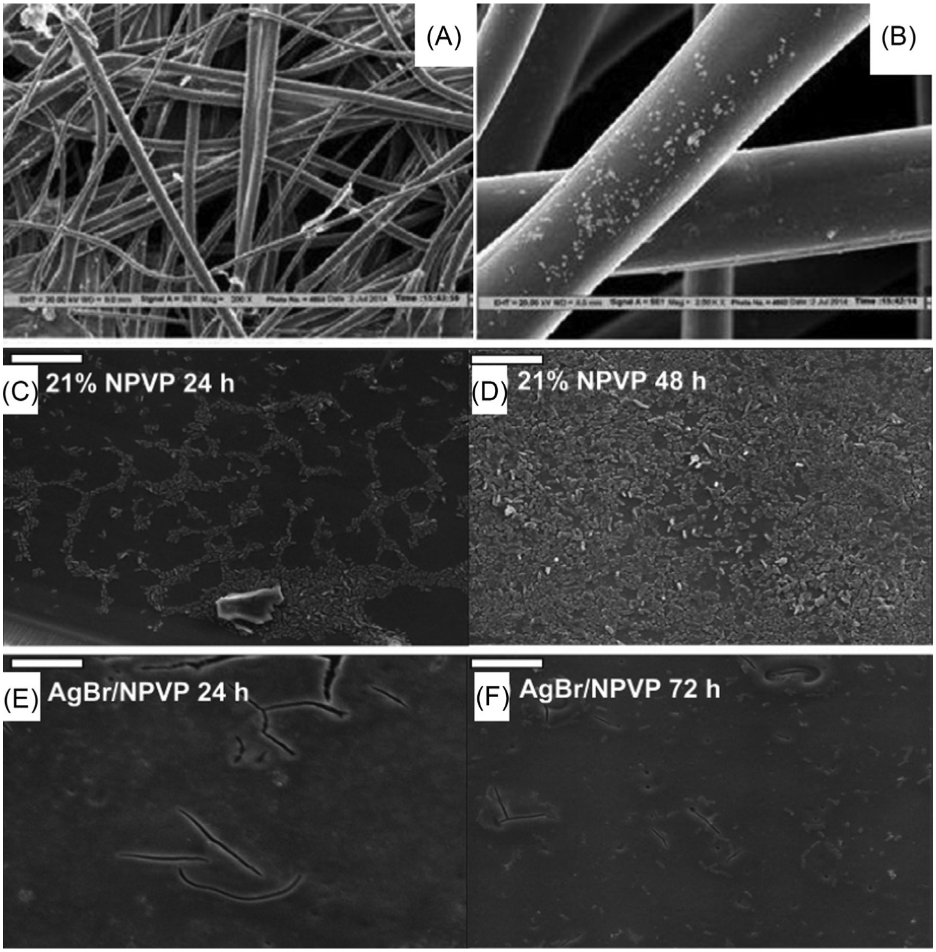

Fig. 14.1 SEM images of (A) and (B) gel/nSnH/Cur. Conditions: 2% gel coated PET fabric. Gel, PVA/PEO/CMC; nSnH, 4.25 μg/cm^2; Cur, 10%; *Aloe vera*, 10%. Reprinted with permission from Anjum, S., Gupta, A., Sharma, D., Gautam, D., Bhan, S., Sharma, A., Kapil, A., Gupta, B., 2016. Development of novel wound care systems based on nanosilver nanohydrogels of polymethacrylic acid with *Aloe vera* and curcumin. Mater. Sci. Eng. C 64, 157–166. Copyright © 2016 Elsevier.SEM image of coated glass surfaces after incubation with *P. aeruginosa*. (C), (D) Biofilm on 21% NPVP-coated glass surfaces after 24 and 48 h incubation. Dense collection of rod-shaped bacteria can be seen colonizing the surface. (E), (F) No biofilm formation observed on 1:1 AgBr/21% NPVP coated glass surfaces even after 72 h incubation. Scale bar is 10 μm. Reprinted with permission from Sambhy, V., MacBride, M.M., Peterson, B.R., Sen, A., 2006. Silver bromide nanoparticle/polymer composites: Dual action tunable antimicrobial materials. J. Am. Chem. Soc. 128, 9798–9808. Copyright © 2006 American Chemical Society.

inhibition. However, it is important to note that various cases of bacterial resistance against Ag are reported in clinical and nonclinical environments. These mechanisms of silver-resistance may be intrinsic or extrinsic. The intrinsic resistance mechanisms include outer membrane permeability, multidrug resistance (MDR) efflux pumps, downregulation of genes, and chromosomal resistance genes. The extrinsic mechanisms include point mutations, adaptive mutations, and plasmids with resistance genes (Salas-Orozco et al., 2019). Thus it is important to reconsider Ag-based nanostructures for antimicrobial applications that rationalize the exploration of alternative bactericidal nanostructures.

14.4.2 Graphene-based nanofilms

Unique surface tuning properties make graphene-based materials extraordinary nanostructures for designing biofilm-resistant antibacterial surfaces. GO and reduced graphene oxide (rGO) nanosheets were the first graphene-based materials reported to have antimicrobial properties. Various stable graphene-polymer dispersions have been fabricated by the incorporation of a biopolymer matrix into graphene to address poor solubility and the tendency of graphene to aggregate due to strong interplaner interactions. The three main strategies to develop GO-polymer nanocomposites are: (1) physical mixing of GO with the polymer, (2) covalent bonding of the polymer matrix via GO functional groups, and (3) attachment of GO with the aromatic group of the polymer via noncovalent interactions such as van der Waals forces, hydrophobic interactions, and π-π stacking. Recently, several biopolymer matrices, such as chitosan, poly(lactic acid) (PLLA), poly-L-lysine (PLL), poly-*N*-vinyl carbazole (PVK), and poly(vinyl alcohol) (PVA), have been reported for the fabrication of graphene-polymer nanocomposites (Kumar et al., 2019).

Graphene oxide quantum dots (GOQDs) were reported to produce antibacterial membranes with reasonably strong antibiofouling properties. The surface modification was carried out using amino-modified PVDF membranes by means of covalent bonding. This covalent bonding led to the fabrication of innovative nano-carbon functionalized membranes having substantially high antibacterial and antibiofouling properties. *E. coli* and *S. aureus* were inactivated up to 89% and 76%, respectively, after a contact period of 1 h. Further, the presence of a GOQD coating layer effectively prevented biofilm formation against *E. coli* and *S. aureus* cells (Fig. 14.2).

The distinctive antimicrobial and antibiofouling performances of GOQDs is attributed to their unique structure and uniform dispersion, enabling the exposure of a larger fraction of active edges and facilitating the formation of oxidation stress. Such GOQD-modified membranes possess long-term stability and durability due to the strong covalent interaction between PVDF and GOQDs (Zeng et al., 2016).

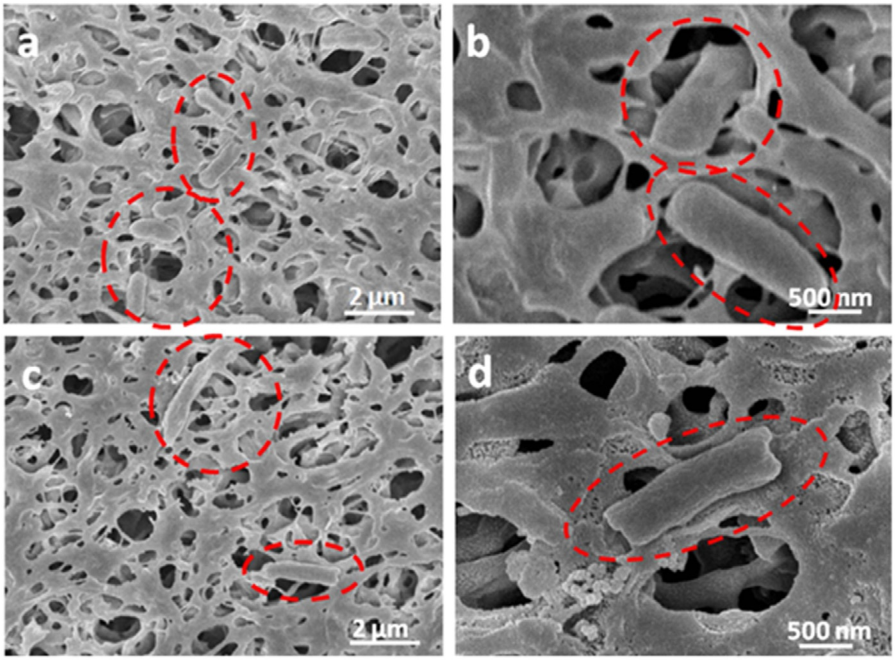

Fig. 14.2 SEM images for normal *E.coli* cells at the surface of a pristine PVDF membrane (A, B) and GOQDs-PVDF membrane (C, D). *(Reprinted with permission from Zeng, Z., Yu, D., He, Z., Liu, J., Xiao, F.X., Zhang, Y., Wang, R., Bhattacharyya, D., Tan, T.T.Y., 2016 Graphene oxide quantum dots covalently functionalized PVDF membrane with significantly-enhanced bactericidal and antibiofouling performances. Sci. Rep. 6, 20142.)*

In another study, graphene nanoplatelets (GNPs) were used as a filler of an experimental dental adhesive for the preparation of an adhesive with antimicrobial and antibiofilm properties. In these polymers, GNPs dispersed uniformly, without agglomeration, and were partially exposed over the adhesive surface. This partial exposure of GNPs is attributed to the antimicrobial and antibiofilm activity against *S. mutans*. Thus, it is evident that both the appropriate porosity of the substrate and rheological properties contributed to the optimal mechanical and antimicrobial properties (Bregnocchi et al., 2017).

14.4.3 Zinc-based nanofilms

ZnONPs have several clinical applications due to their photoluminescence and antioxidant properties. Surface defect-rich ZnONPs have been reported as excellent antidiabetic agents due to their superior α-amylase and α-glucosidase inhibitory potential (Adersh et al., 2015; Kitture et al., 2015). They have also been used as active food packaging agents to prevent the growth of foodborne pathogens. Further, they have been used for the selective targeting of cancer cells since they are known to have limited toxicity towards nontargeted human cells (Al-Ajmi et al., 2015; Zak et al., 2016). Antimicrobial nanocomposite films using a biopolymer based on the fish protein isolate (FPI)/fish skin gelatin (FSG) from fresh yellow stripe trevally (*Selaroides leptolepis*) was prepared at pH 3 and pH 11 containing ZnONPs. The tensile strength (TS) of these nanofilms increased, while elongation at break (EAB) and water vapor permeability (WVP) decreased. FPI/FSG-ZnO nanocomposite films, prepared at pH 3, exhibited strong antibacterial activity against *Listeria monocytogenes*. These results strongly rationalized the use of these biopolymeric nanofilms as potential food packaging materials (Arfat et al., 2016; Arfat et al., 2017).

Biopolymer-like resin filler-based nanocomposites are widely used in medical biology owing to their aesthetic superiority as fillings, strong bonding ability as luting cements, and mechanical reinforcement in restorative procedures. Thus, incorporation of nanostructures like ZnONPs can inhibit biofilm growth on dental composites. Recently, ZnONPs blended at a 10% (*w*/w) fraction with resins resulted in dental composites with antimicrobial and antibiofilm activity. An almost 80% inhibition for a single-species model dental biofilm using *Streptococcus sobrinus* ATCC 27352 was demonstrated (Fig. 14.3). Direct contact inhibition was confirmed to be the underlying mechanism of biofilm inhibition where 20% of the bacterial population survived and could again led to biofilm formation (Sevinc and Hanley, 2010).

Among various polymeric substances, gelatine has been widely used as a thin coating to stabilize surfaces by enlarging the steric barrier, thus gelatin-based coating materials have been explored. Properties of gelatin, like nonimmunogenicity, biodegradability, biocompatibility, and capping for metallic ions, have encouraged the design of various

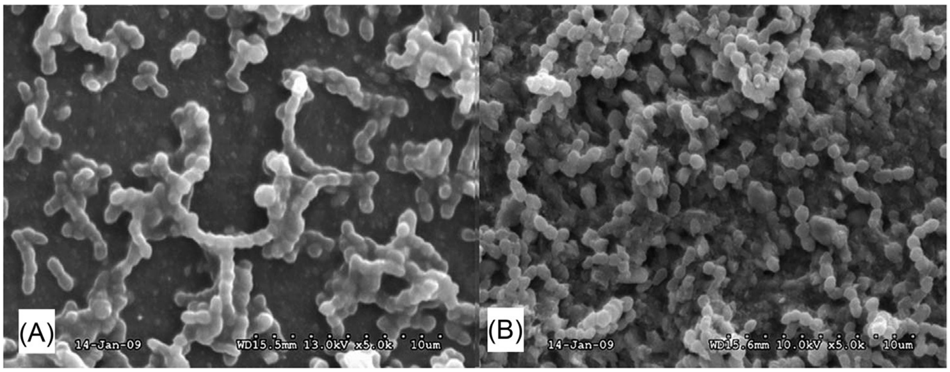

Fig. 14.3 *S. sobrinus* biofilm attachment after 1 day on: (A) 10% ZnONPs and (B) unmodified composites recorded by scanning electron microscopy (×5000 magnification). *(Reprinted with permission from Sevinc, B.A., Hanley, L., 2010. Antibacterial activity of dental composites containing zinc oxide nanoparticles. J. Biomed. Mater. Res. B, Appl. Biomater. 94B, 22–31. Copyright © 2010 Wiley Periodicals, Inc).*

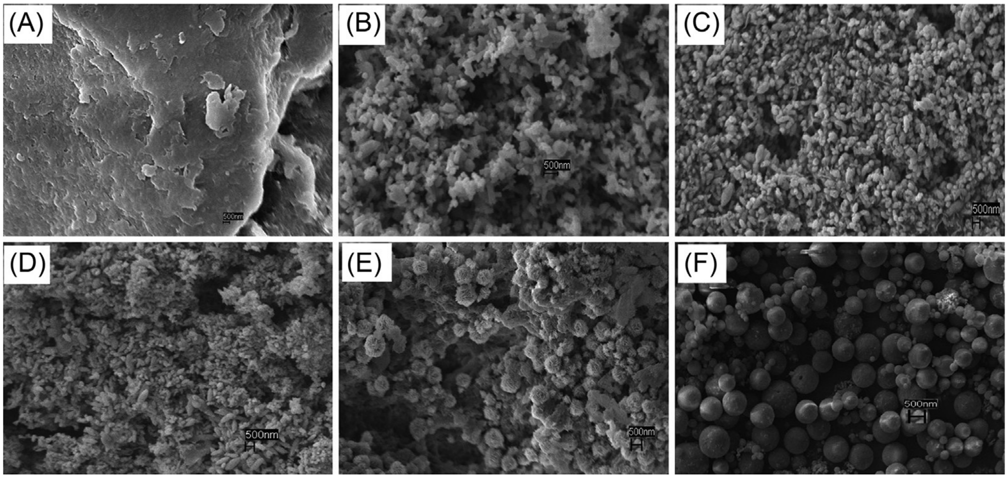

Fig. 14.4 SEM micrographs of: (A) chitosan (powder) and different NPs prepared through a nano spray drying process: (B) ZnONPs; (C) ZnO-CTS; (D) ZnO-CTS-glycerol; (E) ZnO-CTS-starch and; (F) ZnO-CTS-whey powder. All the SEM images were taken at 10 K magnification. *(Reprinted with permission from Dhillon, G.S., Kaur, S., Brar, S.K., 2014, Facile fabrication and characterization of chitosan-based zinc oxide nanoparticles and evaluation of their antimicrobial and antibiofilm activity. Int. Nano Lett. 4, 107. Copyright © 2014).*

bioactive films for biomedical applications (Park et al., 2011).Gelatin-coated ZnONPs (Ge-ZnONPs) synthesized by co-precipitation have shown high antibiofilm potential against Gram-negative bacteria like *Pseudomonas aeruginosa*. Similarly, this novel biomaterial effectively inhibited biofilm growth of the fungus *Candida albicans* at 50 μg/mL, proving Ge-ZnONPs to be promising agents for the control of biofilm-forming microbial pathogens (Divya et al., 2018).

An interesting study reported on the development of chitosan-based ZnONPs using two different methods: nano spray drying and precipitation, using various organic compounds such as citric acid, glycerol, starch, and whey powder as stabilizers (Fig. 14.4). These simple and environmentally benign methods led to the synthesis of antimicrobial surfaces that could resist biofilm formation against *Micrococcus luteus* and *S. aureus* (Dhillon et al., 2014).

14.4.4 Iron-based nanofilms

Mixed matrix membranes (MMMs) are designed using inorganic IONPs in a polymeric matrix and are widely explored for a variety of applications. IONPs are preferred due to their adsorption properties, cost-effectiveness, and nontoxicity (Kitture et al., 2012; Ghosh et al., 2015b, 2019; Kitture and Ghosh, 2019). A Fe_3O_4-polypropylene flat sheet membrane can remove arsenic (v) from water while Fe_3O_4-polyethersulfone-polyanile MMM effectively removes Cu (II). The utility of IONP-integrated polymers was further extended to develop a biofilm-resistant ultrafiltration flat sheet MMM with high flux and adequate antimicrobial properties. This was achieved by mixing IONPs in polyacrylonitrile, resulting in membranes with a molecular weight cut-off in the range of 48 to 65 kDa. *E. coli* was effectively restricted from adhering to such membranes and thus showed a high degree of growth inhibition thereby hindering biofilm formation (Mukherjee and De, 2015).

Dextran [$H(C_6H_{10}O_5)_xOH$]-coated IONPs (DIOM-NPs) have also been used in variety of biomedical applications such as magnetic separation, magnetic resonance imaging, hyperthermia, magnetically guided drug delivery, tissue repair, and molecular diagnostics. DIOM-NPs synthesized by a co-precipitation method were reported to have high anti-adherence activity against *P. aeruginosa* and *E. faecalis* bacterial strains, resulting in the suppression of biofilm formation (Prodan et al., 2014). To combat antibiotic-resistant biofilms, biocompatible multi-compartment nanocarriers composed of both hydrophobic superparamagnetic iron oxide nanoparticles (SPIONs) and the hydrophilic antibiotic methicillin were synthesized. SPION co-encapsulation enhanced both nanocarrier relaxivity and magneticity. It is interesting to note that the iron oxide encapsulating polymersomes (IOPs) could penetrate a 20-μm thick *S. epidermidis* biofilm with high efficiency following the application of an external magnetic field (Fig. 14.5). An optimized IOP formulation comprised of 40 μg/mL SPION and 20 μg/mL of methicillin completely eradicated an entire bacterial population throughout the biofilm. Another remarkable property of this material is its selective toxicity towards methicillin-resistant biofilm-associated bacterial cells but not towards mammalian cells (Geilich et al., 2017).

Chitosan is a polymer of choice for preparing bioactive nano-impregnated polymeric matrices. As such, chitosan-coated IONPs are effective biofilm inhibitors against *S. aureus* on polystyrene surfaces. The proposed reason behind such enhanced activity is that chitosan confers a positive surface charge to the nanoparticles, which enables them to strongly adhere to negatively charged bacteria. Moreover, smaller size chitosan-coated IONPs (15–25 nm) result in better activity compared to the nanoparticles of chitosan alone (200–500 nm). The concentration of chitosan nanoparticles is directly proportional to biofilm inhibition owing to more surface area interacting with bacteria (Shi et al., 2016).

14.4.5 Copper-based nanofilms

Copper nanoparticles (CuNPs) of exotic shapes and desired physico-chemical as well as biological properties have been synthesized using several techniques, such as thermal reduction, capping agent method, sonochemical reduction, metal vapor synthesis, amicroemulsion, laser irradiation, and induced radiation (Singh et al., 2015). However, since these methods are hazardous and expensive, more recently, various medicinal plants like *Dioscorea bulbifera*, *Plumbago zeylanica*, *Gentiana glauca*, *Barleria prionitis*, *Lobelia chinensis*,

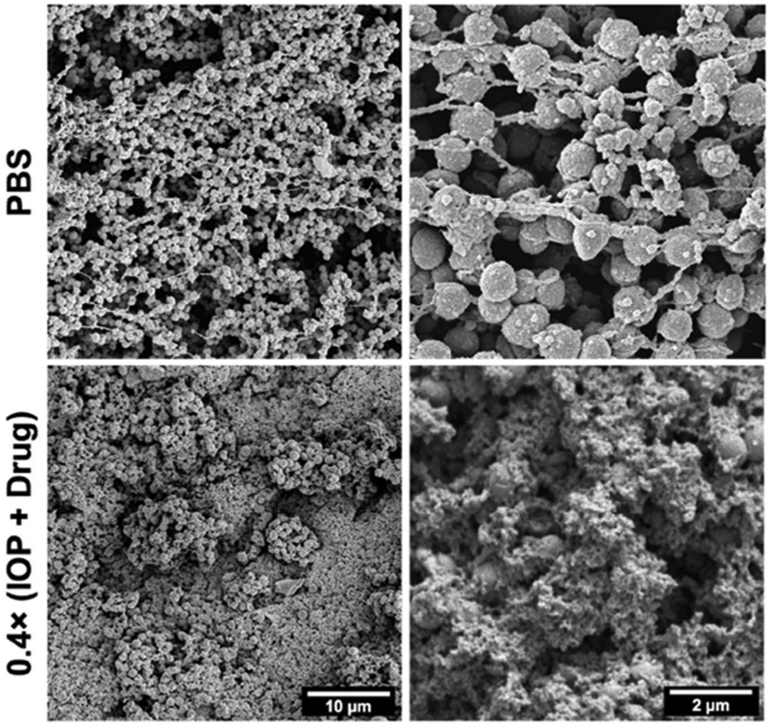

Fig. 14.5 Representative transmission electron micrographs of treated biofilms and a surrounding extracellular polymeric matrix. IOP = iron oxide-encapsulating polymersome. *(Partially reprinted with permission from Geilich, B.M., Gelfat, I., Sridhar, S., van de Ven, A.L., Webster, T.J., 2017. Superparamagnetic iron oxide-encapsulating polymersome nanocarriers for biofilm eradication. Biomaterials 119, 78–85. Copyright© 2016 Elsevier Ltd).*

and *Platanus orientalis* have been used for the synthesis of CuNPs with remarkable antidiabetic and antioxidant properties (Ghosh et al., 2015b; Bhagwat et al., 2018; Jamdade et al., 2019). Similarly, various bacteria (*Enterococcus faecalis, P. fluorescens, P. stutzeri, Morganella morganii, Thermoanaerobacter sp. (X513)* and *Streptomyces sp.*) have been reported to synthesize CuNPs (Ghosh, 2018).

An efficient biopolymer called guar gum (GG) is a hetropolysaccharide of a mannose [(1-4)-linked β-d-mannopyranose] backbone with galactose side groups [(1-6)-linked α-d-galactopyranose] that is extracted from the endosperm of an annual legume plant *Cyamopsis tetragonoloba*. Its high molecular weight, long polymeric chain, wide accessibility, and high solubility provides good film-forming properties. Incorporation of silver-copper alloy nanoparticles (Ag-CuNPs) in GG-based nanocomposite films led to significant alterations in GG's original thermo-mechanical, optical, spectral, oxygen barrier, and antimicrobial properties. Mechanical strength improved while the EAB decreased due to loading of nanoparticles. The GG-based biopolymeric nanofilms exhibited strong antibacterial activity against *L. monocytogenes* and *Salmonella enteric* sv*typhimurium* (Arfat et al., 2017). CuNPs have also been immobilized in a polymeric matrix like polyethyleneimine (PEI), polyacrylonitrile (PAN), and polyethersulfone (PES) to develop antimicrobial membranes (Mukherjee and De, 2015). CuNPs possess broad-spectrum antimicrobial activity against various bacteria including *Klebsiella pneumoniae*, *M. luteus*, *E. coli*, *P. aeruginosa*, *S. aureus*, and *B. subtilis* and inhibit many fungi like *Aspergillus flavus*, *A. niger*, *Alternaria alternata*, *Fusarium solani*, *Penicillium chrysogenum*, and *Candida albicans* (Essa and Khallaf, 2016). Thus, copper-based nanostructures have become one of the best choices for the fabrication of biofilm-resistant surfaces.

Methylacrylate- and ethylmethacrylate-based Primal AC33 polymers and silane/siloxane emulsion-based silicon polymers have been incorporated with CuNPs to design a coating that can prevent biodeterioration of historic monuments and stone works by microbial biofilms. This innovative nanocomposite not only improves the physical and mechanical characters of the treated stones but also significantly inhibits *Streptomyces parvulus*, *E. coli*, and *B. subtilis* (Essa and Khallaf, 2016).

In another study, cotton and polyester were chosen as the substrate material in which CuNPs were uniformly and firmly distributed by the polymer brushes. These reinforced concrete-like structures formed in the composite materials exhibited efficient antibacterial activity against *S. aureus* and *E. coli* even after washing for 30 cycles. This outstanding durability and washable fitness are due to the tethering of CuNPs on the substrates by strong chemical bonds (Sun et al., 2018a,b). An important strategy to resist bio-corrosion in water pipelines is to design a multi-metal oxide nanoparticle-impregnated polymer coating that can protect expensive equipment and reduce maintenance costs. A combination of CuONPs and ZnONPs biologically synthesized using *Sargassum muticum* extract was dispersed in polyethylene oxide polymer and gamma irradiated to reduce their particle size. This nanoparticle polymer composite containing CuONPs and ZnONPs with final particle sizes of 20 and 15 nm, respectively, significantly reduced biofilm formation and adherence of *Proteus mirabilis*, *P. aeruginosa*, and *S. aureus* to carbon mild steel C1010 (Sadek et al., 2019). This composite also inhibited *P. mirabilis*-mediated bio-corrosion on a mild steel coupon by hindering its surface adhesion ability (Fig. 14.6).

Further, wound dressing materials composed of copper-containing nanofiber scaffolds were prepared by the incorporation of copper particles into nanofibers during the electrospinning of poly-D,L-lactide (PDLLA) and poly(ethylene oxide) (PEO). Nanofibers containing copper particles were thinner (326 ± 149 nm in diameter) compared to nanofibers without copper (CF; 445 ± 93 nm in diameter). This composite material could significantly inhibit biofilm formation against *P. aeruginosa* PA01 and *S. aureus* (strain Xen 30) up to 41% and 50%, respectively (Ahire et al., 2016).

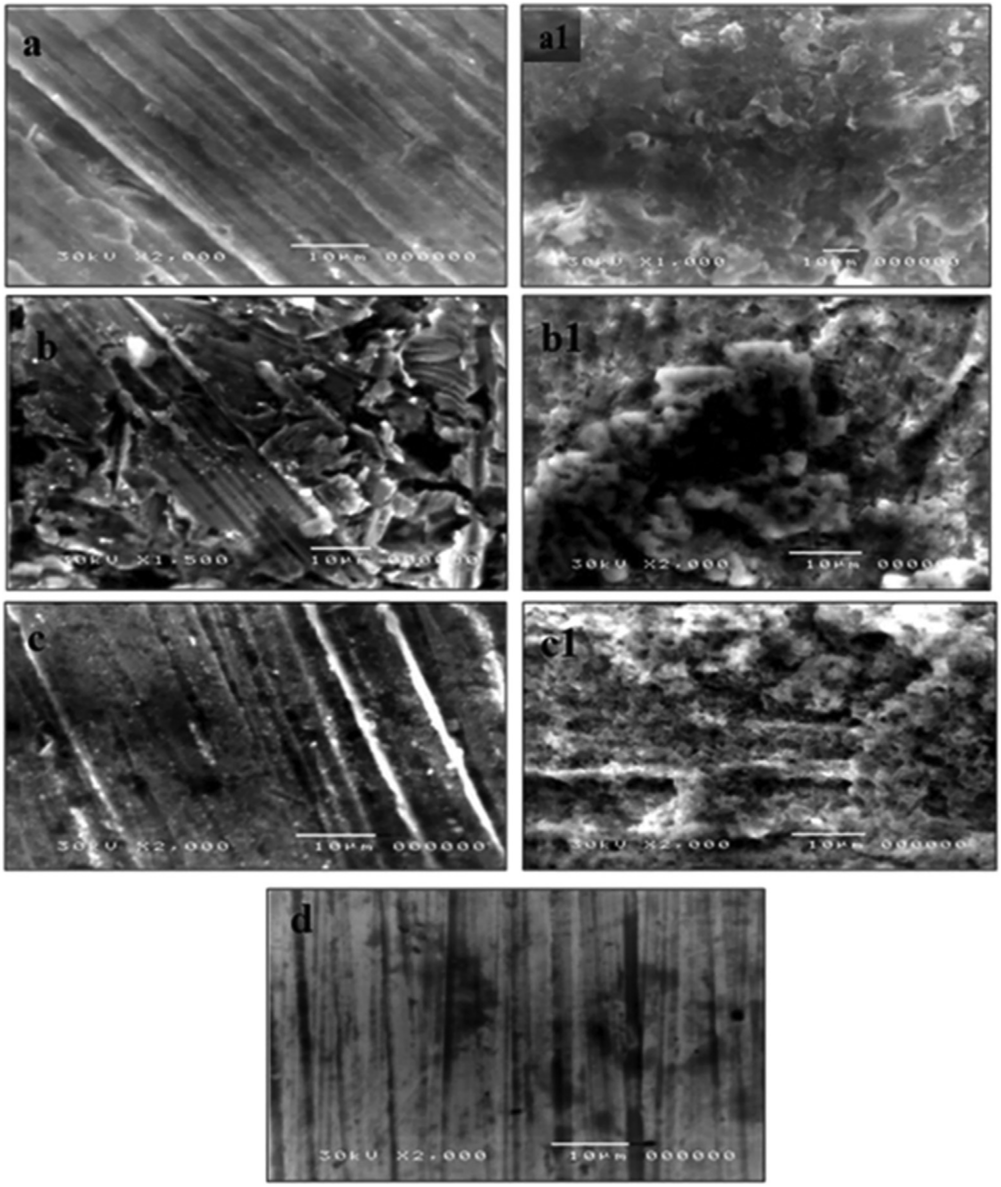

Fig. 14.6 SEM images of coupon's surface after being treated with a CuO/ZnO nanocomposite polymer for 18 h (A), 28 h (B), 48 h (C), and the coupon's surface after embedded in the strongest bio-corrosive strain *Proteus mirabilis* for 18 h (A1), 28 h (B1), 48 h (C1), and (D), control. *(Reprinted with permission from Sadek, R.F., Farrag, H.A., Abdelsalam, S.M., Keiralla, Z., Raafat, A.I., Araby, E., 2019. A powerful nanocomposite polymer prepared from metal oxide nanoparticles synthesized via brown algae as anti-corrosion and antibiofilm. Front. Mater. 6, 140).*

14.4.6 Titanium-based nanofilms

TiO_2NPs are widely known for their photocatalytic purification of water apart from their role in addressing air pollution and developing self-cleaning surfaces (Liu et al., 2017a,b). On irradiation with near-UV light, TiO_2 exhibits strong photocatalytic oxidation, which has been exploited for the removal of trace contaminants and microbial pathogens. Bactericidal effects of TiO_2 on *Lactobacillus acidophilus* rationalize its use in orthodontic appliances, such as pit and fissure sealants, toothbrushes, dental implants, and screws (Huh and Kwon, 2011). A composite comprised of chitosan/starch supplemented nano titanium dioxide (TiO_2-N) and clove oil (CO) and showed remarkable alteration of physico-chemical, biological, and structural properties. Initially, the film-forming dispersion (FFD) was prepared by mixing the chitosan/starch solution, TiO_2-N suspension, and CO emulsion, which was further homogenized at 40°C at 9000 rpm for 5 min. After filtering, the FFD was poured into plastic discs and dried at 38°C under a relative humidity (RH) of 53% (using a saturated sodium bromide solution), which resulted in the formation of the films. Incorporation of TiO_2-N significantly increased the zone of inhibition against *E. coli* and *S. aureus*. Antimicrobial activity in terms of zone diameters could be further enhanced by the addition of CO in the composite. The antibacterial activity of these films was attributed to TiO_2-N and CO-mediated enhancement of non-specific permeability of the cytoplasm membrane eventually leading to the lysis of the cell (Li et al., 2019). TiO_2NPs have been reported to have antibiofilm activity against *S. mitis* ATCC 6249 and Ora-20 (Khan et al., 2016). Similar europium-doped and sulphated anatase nanostructured photocatalysts like TiO_2-S, TiO_2-Eu, and TiO_2-Eu-S showed antibiofilm activity against *E. faecalis* 54 (Dworniczek et al., 2016). Thus, various approaches to synthesize TiO_2NPs integrated polymers have been developed.

An innovative complex polymeric matrix was designed by impregnating an epoxy resin system with Ag^+ ion-doped TiO_2 in order to rationally combine the biocidal characteristics of silver and the photocatalytic properties of TiO_2NPs. This incorporation was achieved by sonicating laboratory-made nanoscale anatase TiO_2 and Ag-TiO_2 into an industrial-grade epoxy resin that resulted in an epoxy composite with 0.5–2.0 wt% of nanofiller content. The composite materials were coated onto a glass petri dish and exhibited significant antibiofilm properties against *S. aureus* ATCC6538 and *E. coli* K-12 due to the combined release of biocide, and photocatalytic activity under static conditions under UV irradiation. Thus, epoxy/Ag-TiO_2 polymer nanocomposite thermoset coating materials may be applied as biofilm-resistant surfaces for industrial or healthcare purposes (Santhosh and Natarajan, 2015). TiO_2NP-functionalized smooth glass surfaces and glass microfiber filters were checked for antibiofilm activity against *S. aureus* and *P. putida*. Irradiation (2 h, 11.2 W m^{-2}, 290–400 nm) led to alterations of a hydrophobic surface to a hydrophilic one. The growth and viability of bacteria greatly reduced due to extensive membrane damage and significant production of intracellular ROS in all

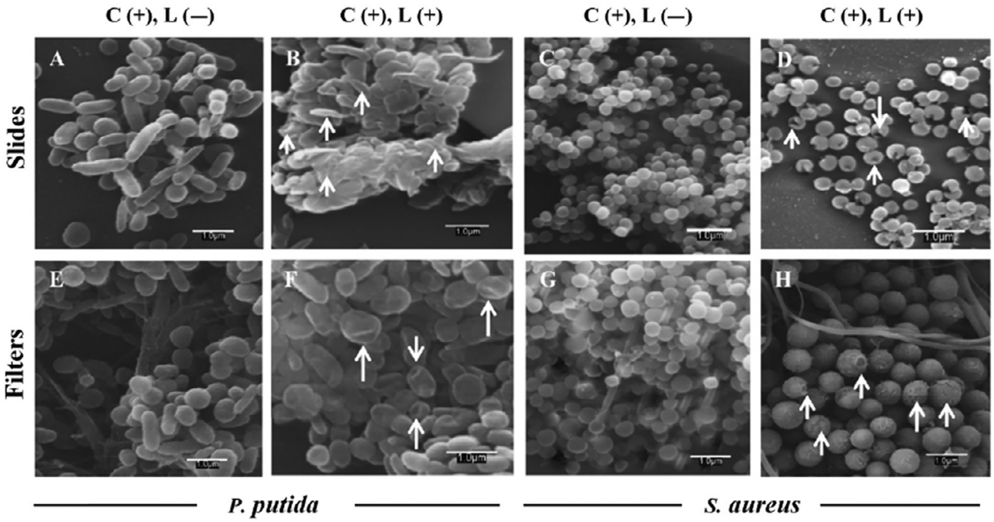

Fig. 14.7 SEM images of *P. putida* (A, B, E and F) and *S. aureus* (C, D, G and H) biofilms on TiO_2-coated glass slides (A, B, C and D) and TiO_2-coated filters (E, F, G and H) before (A, C, E and G) and after irradiation (B, D, F and H). *(Reprinted with permission from Jalvo, B., Faraldos, M., Bahamonde, A., Rosal, R., 2017. Antimicrobial and antibiofilm efficacy of self-cleaning surfaces functionalized by TiO_2 photocatalytic nanoparticles against* Staphylococcus aureus *and* Pseudomonas putida. *J. Hazard. Mater. 340, 160–170. Copyright © 2017 Elsevier B.V).*

TiO_2-loaded irradiated specimens (Fig. 14.7). The reduction of cell viability was more than 99.9% for TiO_2 on glass surfaces, which rationalizes its use in the development of biofim-resistant surfaces (Jalvo et al., 2017).

A composite resin modified by $TiO_2$and TiO_2/AgNPs has also been reported as an attractive antibiofilm filler. In this study, TiO_2 and TiO_2/AgNPs were blended into the composite resin FiltekTMZ350XT at various proportions. These modified composite resins showed excellent antibiofilm activity against *S. mutans* 7-day biofilm, which was found to increase with an increase in the percentage of nanoparticle incorporation. Further, the compressive strength of a composite resin was enhanced due to the treatment of TiO_2NPs with an organosilane, making it an ideal candidate for antibacterial dental restorative material (Dias et al., 2019).

14.4.7 Silica-based nanofilms

Silicone-based materials are widely used in the manufacturing of biomedical devices with an array of applications that include finger and wrist joint reconstruction, breast implants, urinary catheters, and the fabrication of controlled drug delivery devices (such as vaginal rings, intrauterine devices, and transdermal systems) due to their excellent biocompatibility (Depan and Misra, 2014; Ghosh, 2019). However, they are highly prone to microbal adhesion, bacterial colonization, and biofilm formation by various bacterial species such as *Bacillus*, *Enterococcus*, and *Staphylococcus*. Various blended nanofilms have been reported using impregnation of nano-silica into a biopolymeric matrix. Nano silicon dioxide (nano SiO_2) was used to improve the properties of the starch/ PVA films. These biopolymer-blended nanofilms were synthesized by initially mixing 6 g of starch with nano SiO_2 and 4 g of PVA in 100 mL deionized with constant stirring for 5 min. Then, the solution was ultrasonicated for 10 min followed by stirring at 95°C for 20 min. Cross-linking was initiated by the addition of 0.1 g of hexamethylene tetramine, which was again stirred for 40 min followed by the addition of 1.5 g of glycerine as a plasticizer, 0.6 g of tweenum-80 as a surfactant, and 1.2 g of liquid paraffin as a mold release agent. After stirring for 10 min, the mixture was casted onto a glass plate placed on a leveled flat surface and was allowed to dry at 95°C in an oven for 1 h. These blended films with excellent tensile strengths (15.33 MPa) and low water absorption at 37.09% at about 3 wt% of the nano-SiO_2 content exhibited efficient biodegradability (Tang et al., 2009). Wu et al. (2019) reported chitosan films incorporated with curcumin-loaded mesoporous silica nanoparticles (SBA-15) with efficient antimicrobial activity. Initially, curcumin-loaded SBA-15 was prepared by solvent evaporation by adding 100 mg of curcumin dissolved in 5 mL of ethanol to 300 mg of SBA-15, which was further sonicated for 5 min and stirred for 1 h at room temperature in the dark. After drying under nitrogen gas, the unloaded curcumin was removed by dispersing the samples in 10% aqueous ethanol. After washing, the curcumin-loaded mesoporous silica nanoparticles (SBA-15-Cur) were dried at 40°C. The bionanocomposite film was prepared by mixing 1.5% (*w*/w) chitosan in 1% (*v*/v) acetic acid with SBA-15-Cur under stirring at room temperature for 2 h. The resulting dispersion was cast and dried at 50°C for 12 h and further dried at ambient temperature for 2 days. These novel CS/SBA-15-Cur bionanocomposite films showed pH-responsive and sustained release behavior of curcumin. Moreover, enhanced bactericidal activity against both *S. aureus* and *E. coli* strongly rationalized their promising applications as active food packaging materials (Wu et al., 2019).

The high susceptibility of silicone urinary catheters to *E. coli* biofilms is believed to be one of the underlying reasons for urinary tract infections (Stickler and Morgan, 2008). In order to avoid this problem and other associated implant-related infections, a silicone-titania hybrid network elastomer was prepared by the addition of functionalized titania to a silicone base (2 wt% titania, referred to as Silicone-2Ti) by dispersion in an octatrimethylsolixane solvent via sonication in an ice bath and shear mixing for 30 min resulting in a homogenous blend. The enhanced reduction of viability and surface adherence of *S. aureus* was observed due to the rational incorporation of nanophase titania in silicone through a cross-linking mechanism (Fig. 14.8). The underlying mechanism of biofilm disruption of this nanohybrid surface was proposed to be nanoparticle-mediated deformation in the bacterial cell membrane due to puncturing (Depan and Misra, 2014). In order to synthesize biofilm-resistant fillers for dental polymers, silica nanoparticles were

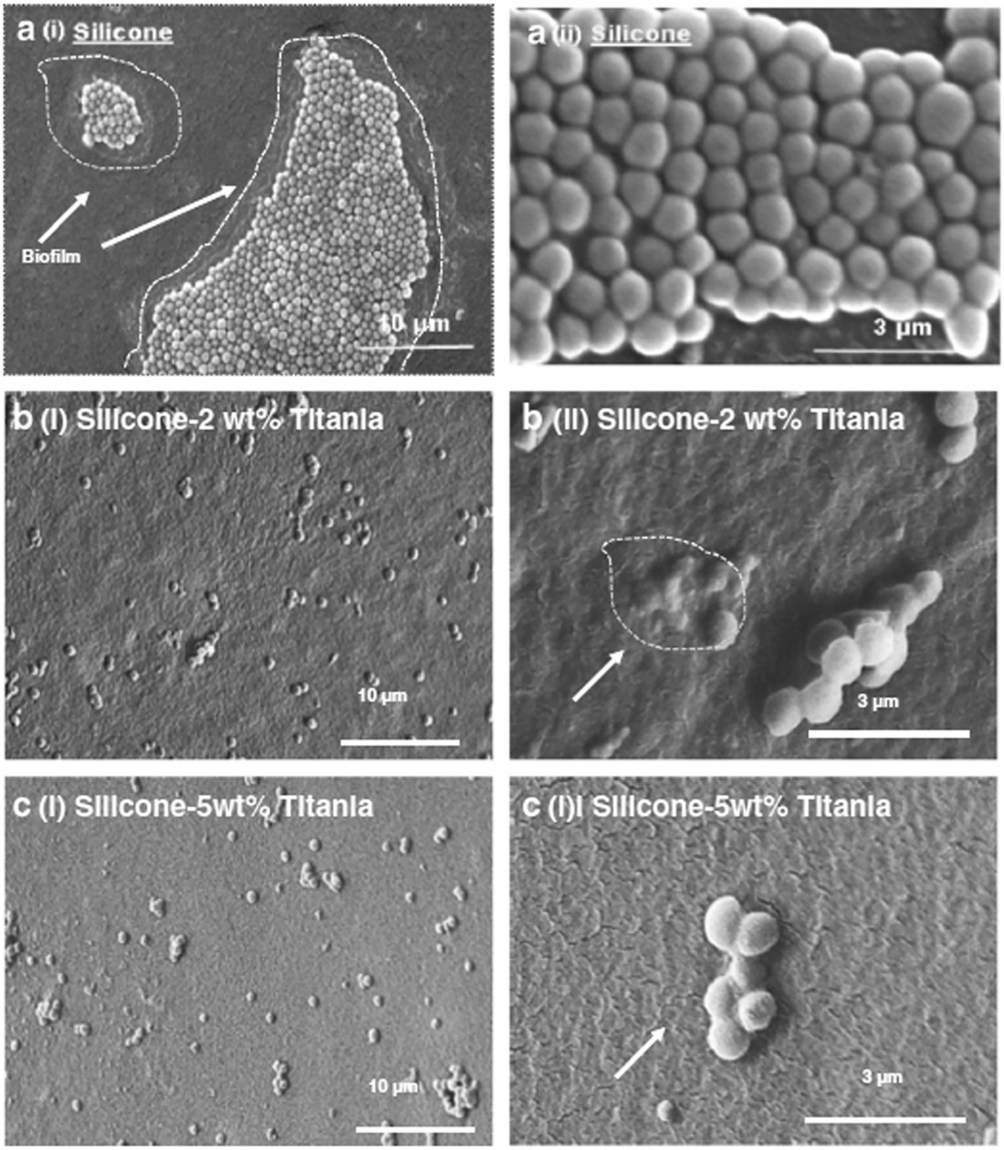

Fig. 14.8 Low magnification *(left)* and high magnification *(right)* scanning electron micrographs illustrating the adhesion and colonization of *S. aureus* after 24 h of incubation on the surface of (A) standalone silicone, (B) silicone-2 wt% titania, and (C) silicone-5 wt% titania. *(Reprinted with permission from Depan, D., Misra, R.D.K., 2014. On the determining role of network structure titania in silicone against bacterial colonization: Mechanism and disruption of biofilm. Mater. Sci. Eng. C 34, 221–228. Copyright © 2013 Elsevier B.V).*

surface modified with: (1) a tertiary amine bearing two t-cinnamaldehyde substituents or (2) dimethyl-octyl ammonium, alongside the well-studied quaternary ammonium polyethyleneimine nanoparticles. These were further incorporated in a commercial dental resin material. Preservation of the degree of conversion and compressive strength of the base materials coupled with enhancement of the surface hydrophobicity was attributed to the addition of cinnamaldehyde-modified particles. Polyethyleneimine particles exhibited the strongest antibacterial effect and cinnamaldehyde-modified particles displayed very high antibiofilm activity against *S. mutans*. Hence, it was confirmed that t-cinnamaldehyde immobilized silica nanoparticles via amine linkers might provide a better biofilm-inhibiting hybrid polymeric surface (Zaltsman et al., 2017).

14.4.8 Zirconia-based nanofilms

Zirconium phosphate ($Zr(HPO_4)_2$) is one of the most useful nanocarriers due to its controllable particle size and intercalation chemistry. These nanostructures can effectively be incorporated with various bioactive agents resulting in a chemically and thermally stable therapeutic agent with enhanced ion exchange capacity, high biocompatibility, and reduced cytotoxicity. Although exotic shapes have been reported, the nano-platelet in particular was found to show better activity. Gels of nanosized ZrP intercalation compounds in aliphatic alcohols prepared by a one-pot synthesis enable intercalation of large cations easily yielding highly efficient ZrP nanoplatelet potential nanocarriers for drug delivery (Ramli and Wong, 2011; Saxena et al., 2013). Various hydrocolloid dressings obtained using sodium carboxymethylcellulose (SCMC) have been widely used in oral and topical pharmaceutical formulations and in wound treatment (Boateng et al., 2008). SCMC films containing chlorhexidine (CLX) loaded into layered zirconium phosphate (ZrP) nanoparticles were reported as promising wound dressing materials with antibiofilm activity but reduced cytotoxicity. Significant biofilm reduction was observed against *S. aureus*, *S. epidermidis*, *S. pyogenes*, and *P. aeruginosa* thereby helping in the treatment of nonhealing and chronic wounds (Donnadio et al., 2016).

14.4.9 Other metal nanostructure-impregnated nanofilms

Although various studies have been conducted on nickel- or molybdenum-based/conjugated compounds, only preliminary studies have been carried out on their nanoparticulate structures for the synthesis of polymer-integrated surfaces. The selection of appropriate biomaterials and nanoparticles is very critical to constructing biofilm-resistant surfaces with biocompatibility. This is achieved by dispersing nickel (Ni), molybdenum (Mo), and tungsten (W) nanoparticles in alkylalkoxysilane (AAS) coatings. Selecting prospective metals like Ni, Mo, and W (which are components of stainless steel and other alloys for biomaterials) serves a dual purpose for biofilm regulation and biocompatibility. NiNP-integrated AAS showed more antibiofilm activity, unlike MoNPs and WNPs, which enhanced biofilm formation. This could be attributed to Ni leakage from the AAS polymer. However, further experimental evidence is necessary to confirm the suitability of this polymer for biomedical applications (Ogawa et al., 2017).

Palladium (Pd) is one of the most bioactive metallic elements reported to produce free radical mediated oxidative stress within the cell, thereby exhibiting high cytotoxicity. Hence, it is used either singly or in combination with other nanostructures, showing a synergistic enhancement in therapeutic index (Rokade et al., 2018, 2017; Ghosh et al., 2015d). Multiwalled carbon nanotubes (MWCNTs) modified with a polypyrrole (PPy) matrix and the incorporation of palladium nanoparticles (PdNPs) with a tube-shaped morphology showed notable biofilm inhibition against *B. subtilis*, *E. coli*, *K. pneumonia*, and *P. aeruginosa* up to 76.72%, 81.71%, 80.96%, and 65%, respectively (Balaji et al., 2018). Another report showed the promise of polysulfone membranes modified by 1,2,3-triazole and PdNPs as potent biofilm inhibitory materials against *P. aeruginosa* in both aerobic and anaerobic conditions. This was further rationalized by the remarkable reduction in total polysaccharides within a monoculture species biofilm matrix. Similarly, a decrease in protein and overall biomass amount in the biofilm layer in the presence of modified membranes endorsed the fact that these modified membranes can act as potent biofouling agents in various industries (Cheng et al., 2016). Gold nanoparticles (AuNPs) of average size 20–27 nm were impregnated in various polymeric matrices like polyurethane (PU), polycaprolactam (PCLm), polycarbonate (PC) and polymethylmethaacrylate (PMMA) by swelling and casting methods and the matrices checked for antibiofilm activity against *E. coli*. Gold-modified surfaces reduced the amount of live bacteria by an order of 2 to 6 and the amount of proteins and carbohydrates reduced by 5 to 12 times on nanocomposites, indicating the superiority of these modified polymeric films over native polymers (Sawant et al., 2013).

14.5 Antimicrobial mechanism behind biofilm control

Integration of nanoparticles in/on polymeric surfaces resist biofilm formation owing to their antimicrobial activity by various means, including alteration of metabolic activity of the pathogen, high penetrating efficiency within biofilms, and more cellular contact mediated interactions due to van der Waals forces and receptor-ligand and hydrophobic interactions. Nanoparticles even cross the bacterial membrane by affecting its integrity, shape, and function. Further, they interact with cellular components like DNA, lysosomes, ribosomes, and enzymes, leading to oxidative stress, heterogeneous alterations, changes in cell membrane permeability, electrolyte balance disorders, enzyme inhibition, protein deactivation, and changes in gene expression (Salunke et al., 2014; Ghosh et al., 2015a). Broadly, various mechanisms of antimicrobial activity of nanoparticles against biofilm-associated and planktonic cells can be classified as in the following sections (Wang et al., 2017).

14.5.1 Oxidative stress

ROS that induce oxidative stress are mainly comprised of the superoxide radical (O_2^-), the hydroxyl radical ($\cdot OH$), hydrogen peroxide (H_2O_2), and singlet oxygen (O_2), which exhibit potent cellular damage in microbial cells. Metal oxides like calcium oxide and magnesium oxide nanoparticles can generate O_2^-, whereas ZnONPs generate H_2O_2 and OH. On the other hand, CuONPs produce all four types of reactive oxygen. $\cdot OH$ and O_2 are more hazardous and can lead to acute microbial death, while O_2^- and H_2O_2 can be neutralized by endogenous antioxidants such as superoxide enzymes and catalase, thus causing reduced oxidative stress. ZnONPs and TiO_2NPs significantly accept light irradiation energy greater than or equal to the band gap stimulating the electrons (e-) in the valence band, which is subjected to transition to the conduction band. Thereafter, this leads to a corresponding hole in the valence band (H^+) resulting in highly reactive reactants (electrons and holes) on the surface of and inside the material. H^+ attaches to the surface of ZnO after interaction with H_2O or OH^-, which is further oxidized to the hydroxyl radical ($\cdot OH$). Likewise, followed by electronic interaction with O_2 and the attachment to the surface of ZnO, the $\cdot OH$ is reduced to O_2^-. The generated ROS interfere with lipid oxidation eventually impairing DNA and RNA synthesis and functioning and degrades the active components that are responsible for maintaining the normal morphological and physiological functions of the microorganism (Yu et al., 2014b; Wang et al., 2017)

14.5.2 Metal ion release

Nanoparticles embedded in a polymeric matrix show an altered metal ion release profile compared to their corresponding metal salts or bulk metal. Metal oxides integrated in the polymeric films show a slow and sustained release of metal ions that target the microbial cells in that vicinity. These metal ions are adsorbed onto the surface of the microbial cell membrane and slowly penetrate the interior of the cell and directly interact with the functional groups of proteins and nucleic acids, such as mercapto (–SH), amino (–NH), and carboxyl (–COOH) groups. This metal ion-biomolecule interaction severely damages various essential enzyme activity, changes the cell structure, affects the normal physiological processes, and ultimately inhibits both biofilm formation as well as planktonic microorganisms. Superparamagnetic iron oxide has been reported to interact with microbial cells by direct penetration within the cell membrane interfering in the transfer of transmembrane electrons (Wang et al., 2017; Hussein-Al-Ali et al., 2014).

14.5.3 Damage to cell wall and cell membranes

Ag^+ ions, apart from deactivating cellular enzymes and DNA by coordination to electron-donating groups such as thiols, carboxylates, amides, imidazoles, indoles, hydroxyls, and so on, are known to cause pits in bacterial cell walls, leading to increased permeability and cell death (Sambhy et al., 2006). AgNPs possess the ability to anchor to the bacterial cell wall and subsequently penetrate it, causing structural changes in the cell membrane. Altered permeability due to cell wall and cell membrane damage leads to leakage of the inner cellular components, eventually leading to cell death. Pit formation increases the accumulation of AgNPs on the cell surface (Palza, 2015). Graphene-based nanoparticles with sharp edges behave as nano-knives, penetrating and disrupting the cell membrane. These 2D nanostructures like GO, reduced-GO (rGO) or GNP wrap the cells inducing mechanical stress and impairing nutrient uptake. Further, they inhibit bacterial adhesion on the substrate (Bregnocchi et al., 2017).

The polycationic charge of chitosan is a key player behind its interaction with teichoic acid binding or extraction of membrane lipids leading to impairment of bacterial cell membranes. It also compromises protein synthesis and restrains microbial growth due to the formation of an external barrier. Further, it has been proposed that chitosan-based nanoparticles may easily enter through water channels present throughout biofilms that are primarily meant for nutrient transportation. This helps the chitosan nanoparticles access the interior of the biofilm matrix, which is inaccessible to other biocidal agents (Shi et al., 2016). One of the very effective modes of photocatalytic membrane damage is due to ROS generation by TiO_2-based nanoparticles upon exposure to near-UV and UVA radiation. The ROS interacts with the components of the cell wall, peptidoglycans (beta-1,4-*N*-acteylglucosamine and beta-1,4-*N*-acetylmuramic acid) and amino acids (e.g., L-alanine, D-glutamine) resulting in lipid peroxidation and formation of malondialdehyde (MDA), which subsequently binds and inactivates cellular proteins and DNA, causing cell death (Depan and Misra, 2014). A cascade of events, like damaging of cell membrane structures, inactivation of DNA/other macromolecules, and oxidation/reduction of the intracellular coenzyme A, finally lead to destruction of the bacterial cells. Photocatalysis-mediated oxidative damage of the cell wall takes place in less than 20 min, followed by the progressive damage of the cytoplasmic membrane and intracellular components. Various other key parameters that play a critical role in the photocatalytic damage of the cells include TiO_2-based nanoparticles and their concentration, type of microorganism, intensity and wavelength of light, degree of hydroxylation, pH, temperature, availability of oxygen, and ROS retention time (Hossain et al., 2014; Liu et al., 2017a,b).

14.5.4 Collapse of proton motive force

The collapse of the proton motive force is another potent mechanism of bactericidal action of ionic silver. Ag^+ ions are released from silver-impregnated membranes, which have been reported to inhibit phosphate uptake and exchange in different bacterial strains (Liu et al., 2017a,b). Ag^+ ions can cause massive damage to the cell boundary by breaking the sophisticated outer membrane and peptidoglycan layer of the cell wall. Eventually, this leads to inactivation of respiratory chain dehydrogenases and ATP synthase enzyme by displacing metallic cations from their binding sites, which is assumed to play a key role in impairment of cellular respiration leading to the death of bacteria (McGivney et al., 2017). Oxidation of fumerate, glucose, and glycerol is disrupted in *E. coli* due to even short-term exposure to AgNPs (Sondi and Salopek-Sondi, 2004). The main reason is due to the interaction between AgNPs and the bacterial membrane, which adversely affects the respiratory chain and cell division. Kim et al. found that Ag^+ ions interact with thiol (–SH) groups of cysteine by replacing the hydrogen atom to form –S–Ag, thereby hindering the enzymatic function of the affected protein to inhibit the growth of *E. coli* (Kim et al., 2012).

14.6 Applications

Bacterial attachment followed by the growth of bacteria on several medical device surfaces, wound dressings, industrial equipment, and textile and food packages is a serious concern worldwide due to the formation of difficult-to-remove entities called biofilms. Thus, the development of antibacterial surfaces that can effectively prevent bacterial adhesion and/or destroy biofilm-forming bacteria (Guo et al., 2013) is highly desirable. Therefore, several antimicrobial agents such as silver, quaternary ammonium moieties, silica-based and carbon-based materials, ROS-generating conjugated polymers, and AMPs have been explored to form an antimicrobial coating on different surfaces to prevent microbial contamination (Chen et al., 2014). Several natural polymers (such as chitosan, gelatin, dextran, alginate, pectin, cellulose, starch, chitin, etc.) have been used for the construction of diverse metal-based antimicrobial nanocomposites denoting their applications in industrial and medical fiels (Zahran and Marei, 2019). Here, we present the application of nanoparticle composite polymeric films as antimicrobial/antibiofilm surfaces in the medical, food, and textile industries.

14.6.1 Medical industry

Medical devices are the most susceptible surfaces for microbial growth and serve as an exceptional medium for hospital-acquired infections. Infected catheters are one of the major sources of nosocomial infections, where bacteria frequently colonize the inner and outer surface of the catheter and form a biofilm. To overcome this, the development of silver-coated impregnated catheters is the most significant contribution in the medical field. Silver has antimicrobial activity for a wide range of medical and industrially important bacteria and is relatively nontoxic to human cells (Dallas et al., 2011). Moreover, an antimicrobial urinary catheter combined with rifampicin, sparfloxacin, and triclosan has been shown to prevent the colonization of *P. mirabilis*, *S. aureus*, and *E. coli* for 7–12 weeks (Fisher et al., 2015).

Moreover, a copolymer of VP-DMMEP (4-vinyl-n-hexylpyridinium bromide (VP) and dimethyl (2-methacryloyloxyethyl) phosphonate) has been developed to reduce biofilm formation on medical devices. It was found that when titanium was coated with a VP-DMMEP copolymer, the adhesion of diverse pathogenic bacteria (i.e. *S. sanguinis*, *E. coli*, and *S. aureus*) was significantly reduced (Winkel et al., 2015). The antimicrobial potential of copper and silver metals at minute concentrations has enhanced their application in diverse fields. Nanoparticles of these metals can be incorporated into polymer matrices to increase their antimicrobial efficacy (Palza, 2015).

Furthermore, a hydrogel containing a dopamine methacrylamide monomer and sodium tetraborate decahydrate embedded with AgNPs posseses antifouling and antibacterial potential and showed great potential for use in wound healing applications (Ghavami Nejad et al., 2016). A wound dressing composed of cotton gauze and incorporated with antimicrobial peptides, polycation, and polyanion showed maximum antimicrobial activity against *S. aureus* and *K. pneumonia* strains. These dressings have also proven to be noncytotoxic to normal human dermal fibroblasts (Gomes et al., 2015). Recently, cationic derivatives of starch polymers containing hydrogels and hydrogel-coated cotton pads have been found to be highly efficient (~100%) against the pathogenic strains of *S. aureus*, *E. coli*, and *P. aeruginosa* (Venkataraman et al., 2019).

Recently, AMPs have attracted the attention of researchers because of their potent antimicrobial activity and negligible antimicrobial resistance. AMPs conjugated into functional polymers such as PEG have application as biomedical nanomaterials. These nanomaterials have maximum antimicrobial potential against a diverse group of pathogenic microorganisms and show high selectivity with low cytotoxicity, suggesting their application as wound dressing, implant coatings, and antibiofilm materials (Sun et al., 2018a,b). Similarly, a β-peptide polymer-modified surface showed excellent antimicrobial activity against drug-resistant bacterial strains, suggesting its use in infection-resistant devices for medical applications (Qi et al., 2019).

Namivandi-Zangeneh et al. (2018) synthesized an antibiofilm polymer containing oligoethylene glycol, ethylhexyl, cationic primary amine, and nitric oxide. This polymer has a dual antimicrobial mechanism, one via the dispersion of a biofilm by releasing nitric oxide and a second through disruption of the bacterial cell wall. In an in vitro study, Ruiz et al. (2019) demonstrated that a biodegradable film made up of chitosan/PVA/GO, when implanted in the subcutaneous tissue of Wistar rats, showed significant antibacterial activity and adequate degradation.

14.6.2 Food industry

The concept of antimicrobial food packaging has expanded enormously due to its ability to increase food safety. Antimicrobial packaging depends on the interaction between food and packaging materials to provide microbial safety and extended shelf life to foodstuffs (Bastarrachea et al., 2011). One of the important advantages of antimicrobial food packaging over the direct addition of antimicrobial agents into foods is packaging's slow and continuous release of antimicrobial agents to food surfaces over a long period of time (Sung et al., 2013).

Chitosan has been widely used as an antimicrobial packaging film due to its potential activities against pathogenic and spoilage microorganisms. Apart from that, chitosan can be molded into fibers, films, gels, sponges, beads, or nanoparticles and it is biodegradable and nontoxic (Dutta et al., 2012). Recently, polyethylene antimicrobial films have been prepared using chitosan and carvacrol-loaded halloysite nanotubes, a tubular clay nanoparticle. This film showed 85% inhibition of the food pathogen *Aeromonas hydrophila* and reduced aerobic cell counts on chicken meat surfaces by 48%. In addition, a coated film surface prevented the attachment of bacterial strains (Tas et al., 2019).

ZnONPs have potential antimicrobial activity and can be incorporated into packaging materials to enhance shelf life and packaging properties of food. Many studies have indicated that paper, glass, and polyurethane coated with ZnONPs have the potential to inhibit a wide range of Gram-positive and Gram-negative bacteria (Espitia et al., 2012). Conversely, a film of low-density polyethylene coated with pyrogallol and polyurethane proved to be effective against *S. aureus* and *E. coli* strains (Gaikwad et al., 2019).

Gomes et al. synthesized poly(D,L-lactide-co-glycolide) (PLGA) nanoparticles with entrapped trans-cinnamaldehyde and eugenol. Eugenol and trans-cinnamaldehyde are natural phytocompounds known to possess potent antimicrobial activity. PLGA NPs have effectively inhibited the growth of *Salmonella* spp. and *Listeria* spp. with concentrations ranging from 20 to 10 mg/mL. Moreover, eugenol and trans-cinnamaldehyde entrapment efficiency released slowly inside the PLGA matrix (Gomes et al., 2011). Moreover, nisin, a widely accepted bacteriocin, was used in a study to develop nisin-loaded chitosan/poly(L-lactic acid) antimicrobial films for food packaging applications. The spontaneous diffusion of nisin from the film demonstrates high antimicrobial activity against *S. aureus* (Wang et al., 2015). A new approach has also been introduced using a cellulose acetate film embedded with a bacteriophage solution. This film showed strong activity against *S. typhimurium* (Gouvêa et al., 2015).

14.6.3 Textile industry

Textiles are favorable substrates for the growth of microorganisms, mostly due to their extremely large surface areas and chemistries. Biofilm-forming bacteria can destroy fabrics, thereby presenting major losses to the textile industry. Consequently, the concept of antimicrobial fabrics has grown over the past decades. The textiles can be loaded with several antimicrobial metal nanoparticles, including palladium, silver, gold, titania, zinc, and so on. Along the same lines, Jang et al. (2015) developed an antimicrobial fiber by loading titania nanorods on mulberry fibers. Similarly, AgNPs have been immobilized on nylon and silk fibers separately by sequential dipping in poly(diallyldimethylammonium chloride) solutions. The AgNP-coated silk fibers and nylon fibers showed an 80% and 50% reduction of *S. aureus*, respectively (Dubas et al., 2006). Similarly, AgNP-treated nylon carpets showed excellent antimicrobial potential with a 99.42% reduction in *S. aureus* and a 79.25% reduction in *E. coli* (Montazer et al., 2012). Moreover, Alonso et al. (2009) reported a significant reduction in the growth of *E. coli* and *Penicillium chrysogenum* by using chitosan cross-linked cellulose fibers as compared to using raw cellulose fibers. Furthermore, ZnONPs and chitosan were used in a study to develop a hybrid antimicrobial coating on cotton fabrics. Antimicrobial activity against pathogenic strains of *S. aureus* and *E. coli* increased significantly by a combination of ZnO and chitosan compared to ZnO or chitosan alone (Petkova et al., 2014). In another study, AgNP-loaded cotton fabrics showed more than a 95% reduction of *E. coli* and *S. aureus* cells without a binder and washing. After a binder was used, the antimicrobial properties were retained even after a 20-wash cycle (Hebeish et al., 2011).

14.7 Conclusions and future perspectives

Nanoparticle-impregnated biopolymeric nanofilms are promising biofilm-resistant and repellent surface coatings to combat nosocomial infections, food spoilage, and microbial colonization on various inanimate surfaces. Biopolymeric membranes have been modified by nanohybrids, individual nanoparticles, and antibiotic or drug-functionalized nanoparticles. Incorporation of such biocidal and antibiofouling agents significantly inhibit the growth and proliferation of pathogens. There are diverse types of bioactive polymers including, hybrid organic-inorganic mixed matrix membranes, nanoparticle-impregnated biocidal polymers, graphene-based films, and functionalized thin film composites. These nanoparticle-immobilized biopolymeric films possess excellent mechanical stability, biofilm resistance, enhanced biocompatibility, and are environmentally friendly, all of which support their applications in pharmaceutical, textile, paint, and food industries. The degree of biofilm inhibition and eradication is highly dependent on the impregnation strategy, concentration of nanoparticles incorporated, surface charge, type of drugs functionalized on the nanoparticles, and structural and genetic make-up of the target pathogen.

Despite the good efficiency of bioactive polymeric films, the emergence of resistance among biofilm-associated microbes residing in the extracellular polymeric secretion (EPS) during long-term exposure remains a major challenge. In order to overcome this problem, strategic inhibition of the initial adhesion of bacteria has been achieved by different approaches such as chemical modification, antibacterial coating, or multi-functionalization by hydrophilic biopolymers.

The production of antibiofim biopolymeric films with inherent self-cleaning and antibiofouling properties has received immense attention for industrial applications.

Biopolymers like chitosan, gelatine, cellulose, starch, and others are preferable for augmentation of nanofilms along with metallic nanoparticles like silica, silver, and zinc, considering their leaching tendency and associated toxicity. Therefore, the rational design of nanomaterial-impregnated biopolymeric films with superior multifaceted properties can promote bacteria contact killing, avoid biofilm formation, and fabricate surfaces with optimum self-disinfecting capability.

Acknowledgments

Dr. Sougata Ghosh acknowledges the Department of Science and Technology (DST), Ministry of Science and Technology, Government of India and Jawaharlal Nehru Centre for Advanced Scientific Research, India for funding under Post-doctoral Overseas Fellowship in Nano Science and Technology (Ref. JNC/AO/A.0610.1(4) 2019-2260 dated August19,2019).

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